Jiancong Xiao

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Academic Experience

National University of Singapore — Singapore
Assistant Professor, Department of Computer Sceince, School of Computing — 2026 - Present

University of Pennsylvania — Philadelphia, PA, USA
Postdoctoral Researcher — 2023 - 2026

- Advisors: Prof. Weijie Su & Prof. Qi Long

Education

The Chinese University of Hong Kong, Shenzhen — Shenzhen, China
Ph.D. in Computer and Information Engineering — 2017 - 2023

- Advisor: Prof. Zhi-Quan (Tom) Luo
- Thesis: Understanding Adversarially Robust Generalization: A Learning Theory Perspective

The Chinese University of Hong Kong — Hong Kong SAR, China
M.Sc. in Mathematics — 2014 - 2015

Sun Yat-sen University — Guangzhou, China
B.Econ. in Finance, Mathematics Honors Program — 2010 - 2014

- Honors program with a dual curriculum in finance and mathematics

Industry Experience

Tencent AI Lab — Shenzhen, China
Research Intern — 2020 - 2022

GF Securities & Futures — Guangzhou, China
Financial Engineer — 2015 - 2017

Publications

(α-β) denotes alphabetical order, * denotes equal contribution, and † denotes corresponding authors.

Journal Articles

[J1] Adversarial Rademacher Complexity of Deep Neural Networks.
Jiancong Xiao, Yanbo Fan, Ruoyu Sun, and Zhi-Quan Luo.
Journal of Machine Learning Research (**JMLR**), 27(140), 1–46, 2026.

[J2] Statistical Impossibility and Possibility of Aligning LLMs with Human Preferences: From Condorcet Paradox to Nash Equilibrium.
(α-β) Kaizhao Liu, Qi Long, Zhekun Shi, Weijie J. Su, and **Jiancong Xiao**.
The Annals of Statistics (**AoS**), 54(4), 2110–2135, 2026.

[J3] On the Algorithmic Bias of Aligning Large Language Models with RLHF: Preference Collapse and Matching Regularization.
Jiancong Xiao, Ziniu Li, Xingyu Xie, Emily Getzen, Cong Fang, Qi Long, and Weijie J. Su.
Journal of the American Statistical Association (**JASA**), 120(552), 2154–2164, 2025.

[J4] Understanding Adversarial Robustness Against On-Manifold Adversarial Examples.
Jiancong Xiao, Liusha Yang, Yanbo Fan, Jue Wang, and Zhi-Quan Luo.
Pattern Recognition (**PR**), 159, 111071, 2025.

Conference Papers

[C1] Beyond Post-Hoc Temperature Scaling: Bilevel Optimization for LLM Calibration.
Ruochen Jin, Zhanliang Wang, Zongyu Dai, **Jiancong Xiao**†, and Bojian Hou†.
The Third Conference on Language Modeling (**COLM**), 2026.

[C2] A Semantic-Sampling Framework for Evaluating Calibration in Open-Ended Question Answering.
Zhanliang Wang*, **Jiancong Xiao***, Ruochen Jin, Shu Yang, Bojian Hou, and Li Shen.
Advances in Neural Information Processing Systems (**NeurIPS**), 39, 2026.

[C3] Restoring Calibration for Aligned Large Language Models: A Calibration-Aware Fine-Tuning Approach.
Jiancong Xiao*, Bojian Hou*, Zhanliang Wang*, Ruochen Jin, Qi Long, Weijie J. Su, and Li Shen.
The 42nd International Conference on Machine Learning (**ICML**), 68364–68390, 2025.

[C4] Fine-Tuning Attention Modules Only: Enhancing Weight Disentanglement in Task Arithmetic.
Ruochen Jin*, Bojian Hou*, **Jiancong Xiao***, Weijie J. Su, and Li Shen.
The Thirteenth International Conference on Learning Representations (**ICLR**), 2025.

[C5] Preserving Diversity in Supervised Fine-tuning of Large Language Models.
Ziniu Li, Congliang Chen, Tian Xu, Zeyu Qin, **Jiancong Xiao**, Zhi-Quan Luo, and Ruoyu Sun.
The Thirteenth International Conference on Learning Representations (**ICLR**), 2025.

[C6] Magnetic Mirror Descent Self-play Preference Optimization.
Mingzhi Wang, Chengdong Ma, Qizhi Chen, Linjian Meng, Yang Han, **Jiancong Xiao**, Zhaowei Zhang, Jing Huo, Weijie J. Su, and Yaodong Yang.
The Thirteenth International Conference on Learning Representations (**ICLR**), 2025.

[C7] Bridging the Gap: Rademacher Complexity in Robust and Standard Generalization.
Jiancong Xiao, Ruoyu Sun, Qi Long, and Weijie J. Su.
The Thirty Seventh Conference on Learning Theory (**COLT**), 5074–5075, 2024.

[C8] Uniformly Stable Algorithms for Adversarial Training and Beyond.
Jiancong Xiao*†, Jiawei Zhang*†, Zhi-Quan Luo, and Asuman E. Ozdaglar.
The 41st International Conference on Machine Learning (**ICML**), 54319–54340, 2024.

[C9] PAC-Bayesian Spectrally-Normalized Bounds for Adversarially Robust Generalization.
Jiancong Xiao, Ruoyu Sun, and Zhi-Quan Luo.
Advances in Neural Information Processing Systems (**NeurIPS**), 36, 36305–36323, 2023.

[C10] Stability Analysis and Generalization Bounds of Adversarial Training.
Jiancong Xiao, Yanbo Fan, Ruoyu Sun, Jue Wang, and Zhi-Quan Luo.
Advances in Neural Information Processing Systems (**NeurIPS Spotlight**), 35, 15446–15459, 2022.

Preprints & Working Papers

[P1] QuantumMind: A Large Chemistry Language Model for Organic Molecular Screening and Discovery.
Yinqi Zeng, **Jiancong Xiao**, Ruizhong Qiu, Yuanchen Bei, Hanghang Tong, and Renjie Li.
Submitted, under review, 2026.

[P2] Theoretical Tensions in RLHF: Reconciling Empirical Success with Inconsistencies in Social Choice Theory.
Jiancong Xiao, Zhekun Shi, Kaizhao Liu, Qi Long, and Weijie J. Su.
Submitted, under review, 2025.

[P3] A Fine-Grained Analysis of Pure Semantic Preference Alignment in Large Language Models.
Ruochen Jin, **Jiancong Xiao**, Nikhil Ruia, Qi Long, and Weijie J. Su.
Submitted, under review, 2025.

[P4] Fundamental Limits of Game-Theoretic LLM Alignment: Smith Consistency and Preference Matching.
Zhekun Shi, Kaizhao Liu, Qi Long†, Weijie J. Su†, and **Jiancong Xiao**†.
Submitted, under review, 2025.

Workshop Papers

[W1] Entropic Distribution Matching in Supervised Fine-tuning of LLMs: Less Overfitting and Better Diversity.
Ziniu Li, Congliang Chen, Tian Xu, Zeyu Qin, **Jiancong Xiao**, Ruoyu Sun, and Zhi-Quan Luo.
NeurIPS 2024 Workshop on Fine-Tuning in Modern Machine Learning: Principles and Scalability (Best Paper Runner-up), 2024.

[W2] Improving Adversarial Training for Multiple Perturbations through the Lens of Uniform Stability.
Jiancong Xiao, Zeyu Qin, Yanbo Fan, Baoyuan Wu, Jue Wang, and Zhi-Quan Luo.
ICML Workshop on New Frontiers in Adversarial Machine Learning, 2023.

[W3] PAC-Bayesian Adversarially Robust Generalization Bounds for Deep Neural Networks.
Jiancong Xiao, Ruoyu Sun, and Zhi-Quan Luo.
ICML Workshop on New Frontiers in Adversarial Machine Learning, 2023.

[W4] Smoothed-SGDmax: A Stability-Inspired Algorithm to Improve Adversarial Generalization.
Jiancong Xiao, Jiawei Zhang, Zhi-Quan Luo, and Asuman E. Ozdaglar.
NeurIPS ML Safety Workshop, 2022.

Talks and Presentation

Invited Talks

- Statistical Foundations of LLM Alignment — Federal Reserve Bank of Kansas City, 2026.05
- Statistical Foundations of LLM Alignment — Shenzhen Loop Area Institute, 2026.02
- LLM Training and Coding for Statistical Learning and Data Science — American Statistical Association, 2025.11
- Statistical Limits of LLM Alignment — Dartmouth College, 2025.10
- Statistical Limits of LLM Alignment — ICSA 2025, University of Connecticut, 2025.06
- Algorithmic Bias of RLHF — Siam MDS 2024, Atlanta, 2024.10
- Algorithmic Bias of RLHF — JSM 2024, Portland, 2024.08
- Adversarial Rademacher Complexity — COLT 2024, University of Alberta, 2024.06
- Stability Analysis of Adversarial Training — IOS 2024, Rice University, 2024.03
- Adversarially Robust Generalization — IOS 2024, Rice University, 2024.03
- Algorithmic Bias of RLHF — University of Pennsylvania, 2023.11
- Uniformly Stable Algorithms of Adversarial Training — MOPTA 2023, Lehigh University, 2023.08
- Stability Analysis of Adversarial Training — Sichuan University, 2023.02
- Adversarial Rademacher Complexity — Sichuan University, 2023.02
- Stability Analysis of Adversarial Training — AI Times Forum, 2022.10
- Metric Entropy — RL Seminar, CUHK(SZ), 2020.08

Poster Presentation

- Restoring Calibration for Aligned Large Language Models — ICML 2025, 2025.07
- Algorithmic Bias of RLHF — 4th Penn Conference on Big Data, University of Pennsylvania, 2024.09
- Algorithmic Bias of RLHF — ASSET Center AI Research Mixer, University of Pennsylvania, 2024.09
- Uniformly Stable Algorithms for Adversarial Training — ICML 2024, 2024.07
- Algorithmic Bias of RLHF — RL workshop, COLT 2024, University of Alberta, 2024.06
- Adversarially Robust Generalization — NeurIPS 2023, New Orleans, 2023.12
- Adaptive Smoothness-Weighted Adversarial Training — AdvML, ICML 2023, Hawaii, 2023.07
- Adversarially Robust Generalization — AdvML, ICML 2023, Hawaii, 2023.07
- Uniformly Stable Algorithms for Adversarial Training — ML Safety, NeurIPS 2022, New Orleans, 2022.12
- Stability Analysis of Adversarial Training — NeurIPS 2022, New Orleans, 2022.12

Awards and Scholarships

- NeurIPS 2026 Top Reviewer — 2026
- NeurIPS 2024 FITML Best Paper Runner-up — 2024
- NeurIPS 2023 Scholar Award — 2023
- NeurIPS 2022 Spotlight — 2022
- Tencent Ph.D. Scholarship — 2020 - 2023
- Poster Award of 2021 Graduate Research Conference at CUHK(SZ) — 2021
- Postgraduate Scholarship by The Chinese University of Hong Kong, Shenzhen — 2017 - 2023
- Undergraduate Scholarship by Sun Yat-sen University — 2013

Teaching

Instructor IVADO, Université de Montréal

- Statistical Foundations of Large Language Models — Summer 2026

Instructor ICSA 2025, University of Connecticut

- Statistical Inference in Large Language Models — Summer 2025

Teaching Assistant CUHK(SZ)

- MAT 3220: Optimization II — Spring 2019 & Spring 2020
- MAT 3007: Optimization I — Fall 2019
- MAT 4001: Numerical Analysis — Fall 2018
- MAT 1004: Mathematical Analysis II — Spring 2018
- MAT 1003: Mathematical Analysis I — Fall 2017

Professional Service

Session Organizer & Chair

- 2024 First Macau International Conference on Business Intelligence and Analytics — University of Macau, 2024.12
  - Session: Optimization and Statistics for Large Language Models
- 37th Annual Conference on Learning Theory (COLT 2024) — University of Alberta, 2024.06
  - Session: Adversarial and Robust Learning
- Modeling and Optimization: Theory and Applications (MOPTA 2023) — Lehigh University, 2023.08
  - Session: Optimization Algorithms of Adversarial Training

Journal Reviewer

- Journal of Machine Learning Research (JMLR).
- IEEE Transactions on Information Theory (TIT).
- Journal of the American Statistical Association (JASA).
- Biometrica.
- Machine Learning.
- Quarterly Journal of Economics and Management (QJEM).

Conference Area Chair

- ICLR 2027.

Conference Reviewer

- Machine Learning: NeurIPS 2022-2026, ICML 2023-2025, ICLR 2023-2026, AISTATS 2022 & 2026.
- Computer Vision: ECCV 2024, CVPR 2023, ICCV 2021.

Languages

- English: Fluent
- Chinese Mandarin and Cantonese: Native